solvesheet

BRAC University (Department of Computer Science and Engineering)

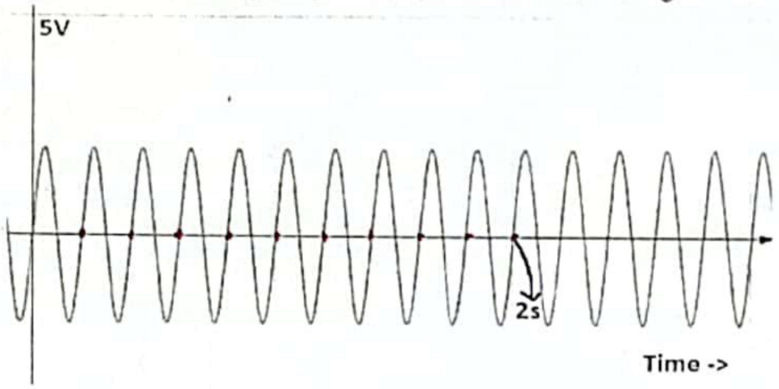
CSE 320 (Data Communication) | Summer 2023

Quiz 2 (Set A)

Full Marks: 10 **Duration: 15 minutes**

Name:	ID:	Section:

1. Determine the (i) Amplitude (ii) Frequency and (iii) Period of the wave given below. **[3 marks]**



Amplitude: 2 V

Frequency: 5 Hz

Period: 0.2 s

2. A sine wave is offset 1/30 cycle with respect to time 0. What is its phase in degrees? **[1 mark]**

Ans: 12

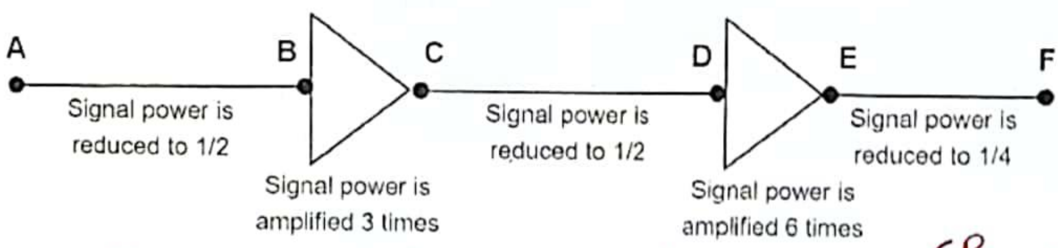
$\frac{1}{30} \times 360$
$= 12$

3. A digital signal has 130 levels. How many bits are needed per level? **[1 mark]**

Ans: 8

$\log_2(130) = 7.0224 \rightarrow 8$

4. Calculate the total change of signal power in decibel and comment if the power is being amplified/attenuated. **[2 marks]**



$$10\log_{10}\frac{\frac{1}{2}P_1}{P_1} + 10\log_{10}\frac{3P_1}{P_1} + 10\log_{10}\frac{\frac{1}{2}P_1}{P_1} + 10\log_{10}\frac{6P_1}{P_1} + 10\log_{10}\frac{\frac{1}{4}P_1}{P_1}$$

$= 0.5115$ dB $\rightarrow$ Amplification

5. Given, a composite periodic signal passing through a channel consists of 8 frequency components of 100, 250, 400, 550, 750, 900, 1050 and 1100 MHz. **[3 marks]**

   a. Draw the spectrum, assuming all components have a maximum amplitude of 10v.

   b. Calculate the bandwidth.

   c. If the signal has 8 levels, what is the maximum bit rate possible assuming the channel is a noiseless one?

a)

b) $B = f_h - f_l = (1100 - 100)\,MHz = 1000\,MHz$

c) $\text{BitRate} = 2 \times 1000 \times \log_2 8$

$= 6000\,Mbps$

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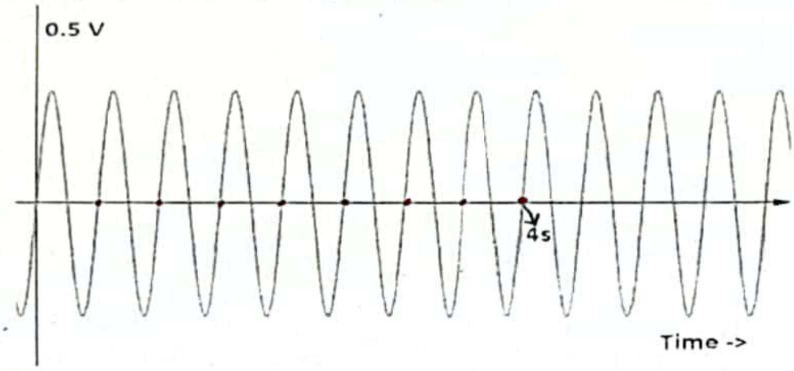
CSE 320 (Data Communication) | Summer 2023

Quiz 2 (Set B)

Full Marks: 10 | **Duration: 15 minutes**

Name:	ID:	Section:

1. Determine the (i) Amplitude (ii) Frequency and (iii) Period of the wave given below. **[3 marks]**



Amplitude: 0.3 V

Frequency: 2 Hz

Period: 0.5 s

2. A sine wave is offset 1/12 cycle with respect to time 0. What is its phase in degrees? **[1 mark]**

Ans: 30

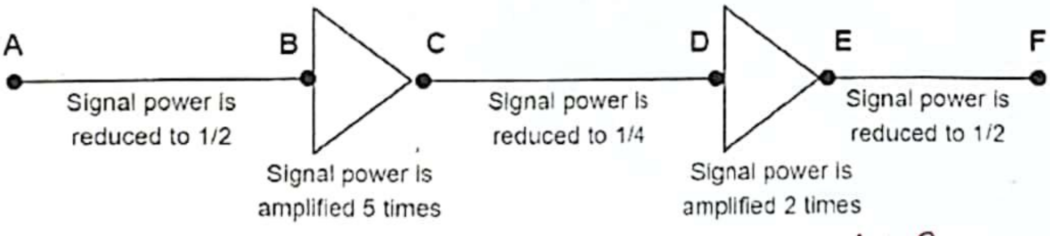
$\frac{1}{12} \times 360 = 30$

3. A digital signal has 260 levels. How many bits are needed per level? **[1 mark]**

Ans: 9

$\log_2(260) = 8.02 \rightarrow 9$

4. Calculate the total change of signal power in decibel and comment if the power is being amplified/attenuated. **[2 marks]**



$$10\log_{10}\frac{1/2\,P_1}{P_1} + 10\log_{10}\frac{5P_1}{P_1} + 10\log_{10}\frac{1/4\,P_1}{P_1} + 10\log_{10}\frac{2P_1}{P_1} + 10\log_{10}\frac{1/2\,P_1}{P_1}$$

$= -2.0412$ dB $\rightarrow$ Attenuation since (-)ve

5. Given, a composite periodic signal passing through a channel consists of 8 frequency components of 150, 300, 500, 650, 850, 1000, 1200 and 1350 MHz. **[3 marks]**
    a. Draw the spectrum, assuming all components have a maximum amplitude of 5v.
    b. Calculate the bandwidth.
    c. If the signal has 16 levels, what is the maximum bit rate possible assuming the channel is a noiseless one?

a)

b) $B = f_h - f_l = (1350 - 150)\,\text{MHz}$

$= 1200\,\text{MHz}$

c) $\text{BitRate} = 2 \times 1200 \times \log_2 16$

$= 9600\,\text{Mbps}$